- an oscillation with frequency 1 in which the two masses move synchronously with the amplitude of motion of the second mass twice that of the first, and
- an oscillation of frequency 2 in which the two masses move in opposite directions with the same amplitude of motion.

Your Turn

You can apply similarly an available computer algebra system to solve Problems 1 through 20 and 39 through 46 in this section (supplying initial conditions for the latter ones if you like).

4.3 Numerical Methods for Systems

We now discuss the numerical approximation of solutions of systems of differential equations. Our goal is to apply the methods of Sections 2.4 through 2.6 to the initial value problem

$$\mathbf{x}' = \mathbf{f}(t, \mathbf{x}), \quad \mathbf{x}(t_0) = \mathbf{x}_0 \tag{1}$$

for a system of m first-order differential equations. In (1) the independent variable is the scalar t, and

$$\mathbf{x} = (x_1, x_2, \ldots, x_m) \quad \text{and} \quad \mathbf{f} = (f_1, f_2, \ldots, f_m)$$

are vector-valued functions. If the component functions of $\mathbf{f}$ and their first-order partial derivatives are all continuous in a neighborhood of the point $(t_0, \mathbf{x}_0)$, then Theorems 3 and 4 of the Appendix guarantee the existence and uniqueness of a solution $\mathbf{x} = \mathbf{x}(t)$ of (1) on some subinterval [of the t-axis] containing t_0. With this assurance we can proceed to discuss the numerical approximation of this solution.

Beginning with step size h, we want to approximate the values of $\mathbf{x}(t)$ at the points $t_1, t_2, t_3, \ldots$, where $t_{n+1} = t_n + h$ for $n \geqq 0$. Suppose that we have already computed the *approximations*

$$\mathbf{x}_1, \quad \mathbf{x}_2, \quad \mathbf{x}_3, \quad \ldots, \quad \mathbf{x}_n$$

to the *actual values*

$$\mathbf{x}(t_1), \quad \mathbf{x}(t_2), \quad \mathbf{x}(t_3), \quad \ldots, \quad \mathbf{x}(t_n)$$

of the exact solution of the system in (1). We can then make the step from $\mathbf{x}_n$ to the next approximation $\mathbf{x}_{n+1} \approx \mathbf{x}(t_{n+1})$ by any one of the methods of Sections 2.4 through 2.6. Essentially all that is required is to write the iterative formula of the selected method in the vector notation of the present discussion.

Euler Method for Systems

For example, the iterative formula of Euler method for systems is

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h\mathbf{f}(t, \mathbf{x}_n). \tag{2}$$

To examine the case $m = 2$ of a pair of first-order differential equations, let us write

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix} \quad \text{and} \quad \mathbf{f} = \begin{bmatrix} f \\ g \end{bmatrix}.$$

Then the initial value problem in (1) is

$$\begin{aligned} x' &= f(t, x, y), \quad x(t_0) = x_0, \\ y' &= g(t, x, y), \quad y(t_0) = y_0, \end{aligned} \tag{3}$$

and the scalar components of the vector formula in (2) are

$$\begin{aligned} x_{n+1} &= x_n + hf(t_n, x_n, y_n), \\ y_{n+1} &= y_n + hg(t_n, x_n, y_n). \end{aligned} \tag{4}$$

Note that each iterative formula in (4) has the form of a single Euler iteration, but with y_n inserted like a parameter in the first formula (for x_{n+1}) and with x_n inserted like a parameter in the second formula (for y_{n+1}). The generalization to the system in (3) of each of the other methods in Sections 2.4 through 2.6 follows a similar pattern.

The improved Euler method for systems consists at each step of calculating first the predictor

$$\mathbf{u}_{n+1} = \mathbf{x}_n + h\mathbf{f}(t_n, \mathbf{x}_n) \tag{5}$$

and then the corrector

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \frac{h}{2}[\mathbf{f}(t_n, \mathbf{x}_n) + \mathbf{f}(t_{n+1}, \mathbf{u}_{n+1})]. \tag{6}$$

For the case of the two-dimensional initial value problem in (3), the scalar components of the formulas in (5) and (6) are

$$\begin{aligned} u_{n+1} &= x_n + hf(t_n, x_n, y_n), \\ v_{n+1} &= y_n + hg(t_n, x_n, y_n) \end{aligned} \tag{7}$$

and

$$\begin{aligned} x_{n+1} &= x_n + \frac{h}{2}[f(t_n, x_n, y_n) + f(t_{n+1}, u_{n+1}, v_{n+1})], \\ y_{n+1} &= y_n + \frac{h}{2}[g(t_n, x_n, y_n) + g(t_{n+1}, u_{n+1}, v_{n+1})]. \end{aligned} \tag{8}$$

Example 1 Consider the initial value problem

$$\begin{aligned} x' &= 3x - 2y, \qquad x(0) = 3; \\ y' &= 5x - 4y, \qquad y(0) = 6. \end{aligned} \tag{9}$$

The exact solution of the system in (9) is

$$x(t) = 2e^{-2t} + e^t, \qquad y(t) = 5e^{-2t} + e^t. \tag{10}$$

Here we have $f(x, y) = 3x - 2y$ and $g(x, y) = 5x - 4y$ in (3), so the Euler iterative formulas in (4) are

$$x_{n+1} = x_n + h \cdot (3x_n - 2y_n), \quad y_{n+1} = y_n + h \cdot (5x_n - 4y_n).$$

With step size $h = 0.1$ we calculate

$$\begin{aligned} x_1 &= 3 + (0.1) \cdot [3 \cdot 3 - 2 \cdot 6] = 2.7, \\ y_1 &= 6 + (0.1) \cdot [5 \cdot 3 - 4 \cdot 6] = 5.1 \end{aligned}$$

and

$$\begin{aligned} x_2 &= 2.7 + (0.1) \cdot [3 \cdot (2.7) - 2 \cdot (5.1)] = 2.49, \\ y_2 &= 5.1 + (0.1) \cdot [5 \cdot (2.7) - 4 \cdot (5.1)] = 4.41. \end{aligned}$$

The actual values at $t_2 = 0.2$ given by (10) are $x(0.2) \approx 2.562$ and $y(0.2) \approx 4.573$.

To compute the improved Euler approximations to $x(0.2)$ and $y(0.2)$ with a single step of size $h = 0.2$, we first calculate the predictors

$$u_1 = 3 + (0.2) \cdot [3 \cdot 3 - 2 \cdot 6] = 2.4,$$
$$v_1 = 6 + (0.2) \cdot [5 \cdot 3 - 4 \cdot 6] = 4.2.$$

Then the corrector formulas in (8) yield

$$x_1 = 3 + (0.1) \cdot ([3 \cdot 3 - 2 \cdot 6] + [3 \cdot (2.4) - 2 \cdot (4.2)]) = 2.58,$$
$$y_1 = 6 + (0.1) \cdot ([5 \cdot 3 - 4 \cdot 6] + [5 \cdot (2.4) - 4 \cdot (4.2)]) = 4.62.$$

As we would expect, a single improved Euler step gives better accuracy than two ordinary Euler steps. ■

The Runge–Kutta Method and Second-Order Equations

The vector version of the iterative formula for the Runge–Kutta method is

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4), \tag{11}$$

where the vectors $\mathbf{k}_1$, $\mathbf{k}_2$, $\mathbf{k}_3$, and $\mathbf{k}_4$ are defined (by analogy with Eqs. (5a)–(5d) of Section 2.6) as follows:

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{f}(t_n, \mathbf{x}_n), \\ \mathbf{k}_2 &= \mathbf{f}\left(t_n + \tfrac{1}{2}h, \mathbf{x}_n + \tfrac{1}{2}h\mathbf{k}_1\right), \\ \mathbf{k}_3 &= \mathbf{f}\left(t_n + \tfrac{1}{2}h, \mathbf{x}_n + \tfrac{1}{2}h\mathbf{k}_2\right), \\ \mathbf{k}_4 &= \mathbf{f}(t_n + h, \mathbf{x}_n + h\mathbf{k}_3). \end{aligned} \tag{12}$$

To describe in scalar notation the Runge–Kutta method for the two-dimensional initial value problem

$$\begin{aligned} x' &= f(t, x, y), \qquad x(t_0) = x_0, \\ y' &= g(t, x, y), \qquad y(t_0) = y_0, \end{aligned} \tag{3}$$

let us write

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} f \\ g \end{bmatrix}, \quad \text{and} \quad \mathbf{k}_i = \begin{bmatrix} F_i \\ G_i \end{bmatrix}.$$

Then the Runge–Kutta iterative formulas for the step from (x_n, y_n) to the next approximation $(x_{n+1}, y_{n+1}) \approx (x(t_{n+1}), y(t_{n+1}))$ are

$$\begin{aligned} x_{n+1} &= x_n + \frac{h}{6}(F_1 + 2F_2 + 2F_3 + F_4), \\ y_{n+1} &= y_n + \frac{h}{6}(G_1 + 2G_2 + 2G_3 + G_4), \end{aligned} \tag{13}$$

where the values F_1, F_2, F_3, and F_4 of the function f are

$$\begin{aligned} F_1 &= f(t_n, x_n, y_n), \\ F_2 &= f\left(t_n + \tfrac{1}{2}h, x_n + \tfrac{1}{2}hF_1, y_n + \tfrac{1}{2}hG_1\right), \\ F_3 &= f\left(t_n + \tfrac{1}{2}h, x_n + \tfrac{1}{2}hF_2, y_n + \tfrac{1}{2}hG_2\right), \\ F_4 &= f(t_n + h, x_n + hF_3, y_n + hG_3); \end{aligned} \tag{14}$$

Note that both F_1 and G_1 must be computed before F_2 and G_2 can be found, etc.

t	$x = \sin t$	$y = \cos t$
0.5	+0.47943	+0.87758
1.0	+0.84147	+0.54030
1.5	+0.99749	+0.07074
2.0	+0.90930	−0.41615
2.5	+0.59847	−0.80114
3.0	+0.14112	−0.98999
3.5	−0.35078	−0.93646
4.0	−0.75680	−0.65364
4.5	−0.97753	−0.21080
5.0	−0.95892	+0.28366

FIGURE 4.3.1. Runge–Kutta values (with $h = 0.05$) for the problem in Eq. (18).

G_1, G_2, G_3, and G_4 are the similarly defined values of the function g.

Perhaps the most common application of the two-dimensional Runge–Kutta method is to the numerical solution of second-order initial value problems of the form

$$\begin{aligned} x'' &= g(t, x, x'), \\ x(t_0) &= x_0, \quad x'(t_0) = y_0. \end{aligned} \tag{15}$$

If we introduce the auxiliary variable $y = x'$, then the problem in (15) translates into the two-dimensional first-order problem

$$\begin{aligned} x' &= y, & x(t_0) &= x_0, \\ y' &= g(t, x, y), & y(t_0) &= y_0. \end{aligned} \tag{16}$$

This is a problem of the form in (3) with $f(t, x, y) = y$.

If the functions f and g are not too complicated, then it is feasible to carry out manually a reasonable number of steps of the two-dimensional Runge–Kutta method described here. But the first operating electronic computers were constructed (during World War II) specifically to implement methods similar to the Runge–Kutta method for the numerical computation of trajectories of artillery projectiles. The application material for this section lists TI-Nspire™ CX II CAS and BASIC versions of Program RK2DIM that can be used with two-dimensional systems.

Example 2

The exact solution of the initial value problem

$$x'' = -x; \quad x(0) = 0, \quad x'(0) = 1 \tag{17}$$

is $x(t) = \sin t$. The substitution $y = x'$ translates (17) into the two-dimensional problem

$$\begin{aligned} x' &= y, & x(0) &= 0; \\ y' &= -x, & y(0) &= 1, \end{aligned} \tag{18}$$

which has the form in (3) with $f(t, x, y) = y$ and $g(t, x, y) = -x$. The table in Fig. 4.3.1 shows the results produced for $0 \leqq t \leqq 5$ (radians) using Program RK2DIM with step size $h = 0.05$. The values shown for $x = \sin t$ and $y = \cos t$ are all accurate to five decimal places. ■

Example 3

Lunar lander In Example 4 of Section 2.3 we considered a lunar lander that initially is falling freely toward the surface of the moon. Its retrorockets, when fired, provide a deceleration of $T = 4$ m/s^2. We found that a soft touchdown at the lunar surface is achieved by igniting these retrorockets when the lander is at a height of 41,870 meters (just over 26 miles) above the surface and is then descending at the rate of 450 m/s.

Now we want to compute the *descent time* of the lunar lander. Let the distance $x(t)$ of the lander from the center of the moon be measured in meters and measure time t in seconds. According to the analysis in Section 2.3 (where we used $r(t)$ instead of $x(t)$), $x(t)$ satisfies the initial value problem

$$\begin{aligned} &\frac{d^2x}{dt^2} = T - \frac{GM}{x^2} = 4 - \frac{4.9044 \times 10^{12}}{x^2}, \\ &x(0) = R + 41870 = 1{,}781{,}870, \quad x'(0) = -450 \end{aligned} \tag{19}$$

where $G \approx 6.6726 \times 10^{-11}$ N·(m/kg)^2 is the universal gravitational constant and $M = 7.35 \times 10^{22}$ kg and $R = 1.74 \times 10^6$ m are the mass and radius of the moon. We seek the value of t when $x(t) = R = 1{,}740{,}000$.

The problem in (19) is equivalent to the first-order system

$$\begin{aligned} \frac{dx}{dt} &= y, \quad x(0) = 1{,}781{,}870; \\ \frac{dy}{dt} &= 4 - \frac{4.9044 \times 10^{12}}{x^2}, \quad y(0) = -450. \end{aligned} \tag{20}$$

The table in Fig. 4.3.2 shows the result of a Runge–Kutta approximation with step size $h = 1$ (the indicated data agreeing with those obtained with step size $h = 2$). Evidently, touchdown on the lunar surface ($x = 1{,}740{,}000$) occurs at some time between $t = 180$ and $t = 190$ seconds. The table in Fig. 4.3.3 shows a second Runge–Kutta approximation with $t(0) = 180$, $x(0) = 1{,}740{,}059$, $y(0) = -16.83$, and $h = 0.1$. Now it is apparent that the lander's time of descent to the lunar surface is very close to 187 seconds; that is, 3 min 7 s. (The final velocity terms in these two tables are positive because the lander would begin to ascend if its retrorockets were not turned off at touchdown.) ■

t (s)	x (m)	v (m/s)
0	1,781,870	−450.00
20	1,773,360	−401.04
40	1,765,826	−352.37
60	1,759,264	−303.95
80	1,753,667	−255.74
100	1,749,033	−207.73
120	1,745,357	−159.86
140	1,742,637	−112.11
160	1,740,872	−64.45
180	1,740,059	−16.83
200	1,740,199	30.77

FIGURE 4.3.2. The lander's descent to the lunar surface.

t (s)	x (m)	v (m/s)
180	1,740,059	−16.83
181	1,740,044	−14.45
182	1,740,030	−12.07
183	1,740,019	−9.69
184	1,740,011	−7.31
185	1,740,005	−4.93
186	1,740,001	−2.55
187	1,740,000	−0.17
188	1,740,001	2.21
189	1,740,004	4.59
190	1,740,010	6.97

FIGURE 4.3.3. Focusing on the lunar lander's soft touchdown.

Higher-Order Systems

As we saw in Section 4.1, any system of higher-order differential equations can be replaced with an equivalent system of first-order differential equations. For example, consider the system

$$\begin{aligned} x'' &= F(t, x, y, x', y'), \\ y'' &= G(t, x, y, x', y') \end{aligned} \tag{21}$$

of second-order equations. If we substitute

$$x = x_1, \quad y = x_2, \quad x' = x_3 = x_1', \quad y' = x_4 = x_2',$$

then we get the equivalent system

$$\begin{aligned} x_1' &= x_3, \\ x_2' &= x_4, \\ x_3' &= F(t, x_1, x_2, x_3, x_4), \\ x_4' &= G(t, x_1, x_2, x_3, x_4) \end{aligned} \tag{22}$$